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Customized Plugging Strength: An Advanced Technique for Chemical Water Shut-Off in Viscous Crude Reservoirs

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Abstract

High water cut is a major challenge affecting the development of heavy oil reservoirs, particularly prominent in thermal recovery. Multiple factors such as reservoir development duration, heterogeneity, oil-water flow ratio, well pattern type, and production parameters influence water cut rise, and the control technologies must be comprehensively consideredfor these factors. How to accurately determine and select appropriate water shut-off strategies is a key issue for the efficient development of heavy oil fields. Firstly, it used the independently developed Intelligent Diagnosis Technology for Water Flooding Status in Complex Reservoirs (FLIIS™) to quantitatively identify and calculate dominant seepage channels. Secondly, it optimized customized plugging strengths and developed different chemical agents such as thermosensitive reversible gels and high-temperature-resistant foaming agents.At last it analyzed water invasion patterns by combining water breakthrough time and water cut rising rate to construct a step-by-step plugging design workflow. The thermosensitive reversible gel was an intelligent responsive gel system, which existed as a low-viscosity fluid below 80°C and forms a colloid above 80°C. After gelation, its viscosity exceeded30,000 mPa·s, and the gel strength reached grade H, which can withstand temperatures above 180°C. The plugging rate exceeded 99.7% after core injection and gelation, indicating its strong plugging ability under thermal recovery conditions. Meanwhile, the product formula can be adjusted according to different reservoir temperatures to better adapt to the reservoir. The thermo-stable foaming agent had the properties of high temperature and salt resistance, with a temperature resistance of up to 350°C. Its resistance factor at 250°C was greater than 30, which can inhibit gas channeling in multiple cycles of thermal recovery and improve development effects. Field applications show that in the first-round measures for Well A1, the initial water cut decreased from 74% to below 5%, the peak daily oil production reached 53.9 m³, cumulative oil

increase was 4,100 m^3, and the effective period lasted approximately 5 months. The thermosensitive gel alleviated inter-well gas channeling interference during single-well heat injection, increasing oil production by 40%-60% after thermal recovery.

Keywords: high water cut, heavy oil, water shut off, chemical agent, new strategy

Introduction

Heavy oil, known as viscous crude, is internationally defined as a kind of crude oil with an in-situ viscosity exceeding 50 mPa·s, usually which typically exhibits a high density. According to data released by the Energy Minerals Division of the American Association of Petroleum Geologists (AAPG) in 2019, global bitumen and heavy oil resources were estimated at approximately 9.38×10^{11} tons, with over 80% located in Canada, Venezuela, and the United States[1]. China ranked as the world's fourth-largest holder of heavy oil resources, having discovered over 70 heavy oil fields across more than ten basins. At present, theprimary viscosity reduction technologies employed domestically and internationally include steam injection, in-situ combustion, and hybrid thermal methods[2-3].

Globally, offshore heavy oil reserves were primarily located in the Gulf of Mexico, North Sea, Mediterranean Sea, and Congo Zatchi regions. However, development in these areas predominantly relied on cold production methods like waterflooding and chemical flooding, with no precedent for large-scale thermal recovery operations from offshore platforms. Offshore heavy oil resources in China are abundant, making thermal recovery a crucial strategy for increasing offshore production. China's Bohai Bay hosted significant offshore heavy oil reserves, accounting for 70-85% of the Original Oil in Place (OOIP). Among these, reserves totaling 2.4669 billion cubic meters exhibit viscosities exceeding 400 cP, establishing heavy oil as a vital component of offshore production[4]. Since initiating Multi-Component Thermal Fluid (MCTF) huff-and-puff trials in 2008, China National Offshore Oil Corporation (CNOOC) had developed a preliminary offshore heavy oil thermal recovery technology system through over a decade of research and field application. Implementing a strategy of "pilot testing, technology demonstration, and scaled application," CNOOC has successfully achieved large-scale thermal development in fields such as NB-1, LD-1, and LD-2. By 2024, offshore heavy oil thermal production exceeded 1 million tons[5].

The unique offshore operating environment, characterized by "large well spacing, long-period, and limited space," presented challenges such as high economic thresholds and stringent safety requirements. Offshore injection production well spacing was 300-400 m,which was significantly larger than onshore spacing by 70-150 m. Similarly, the cyclic heat injection of offshore oilfield was 6000-8000t, and that of onshore oilfield was 1500-2000t.Moreover, the cumulative production for offshore well was 120,000-160,000 tons, which was typically 5-6 times higher than onshore well[6]. Compared to onshore fields, the high porosity and permeability of offshore reservoirs, combined with high-intensity steam injection, significantly influenced fluid flow characteristics in porous media, the formation and propagation of channeling pathways, and overall production performance. Research results indicated that increasing injection intensity leads to a sequential processin the expansion of the heated chamber, including dilation, fracturing,formation of dominant flow channels, and pressure release, ultimately resulting in the development of channeling pathways[7].

As large-scale thermal recovery progresses in the Bohai offshore region, challenges such as inter-well steam channeling and edge and bottom water breakthrough had emerged. Newly discovered oilfields exhibited more complexproblems, such as increasingly complex oil-water systems, multi-layer heavy oil reservoirs and thin oil columns underlain by bottom water. Based on enhanced geological and reservoir understanding, future targets for offshore thermal stimulation will expand from single sand bodies to multi-oil-water-system thick layers, also from conventional heavy oil to extraheavy or ultra-heavy oil, and from edgewater to edge-bottom-topwater reservoirs. The increasingcomplexity will lead to more severe problems, including top or bottom water breakthrough, reduced steam sweep efficiency, and poor conformance. Therefore, it is imperative to develop high-temperature resistant, high-strength, and environmentally friendly plugging systems for water shut-off and channeling control,which is essential to enlarge the steam swept volume, enhance heating efficiency, improve the economic viability of offshore heavy oil development, and advance offshore thermal recovery technologies.

Methodology

The core principle of this technology centered on the identification of channeling pathways and the customization of water shut-off agents, integrated with optimized process parameter implementation and dynamic control techniques. It achieved precise water shut-off through the identification and detailed characterization of thermal recovery channeling pathways, ultimately establishing a closed-loop management system of "Design-Implementation-Evaluation-Optimization." Key breakthroughs were required in addressing challenges such as the stability and plugging selectivity of water control agents under high-temperature, high-salinity conditions to promote the efficient development of heavy oil reservoirs.

Step 1: identification and detailed characterization of thermal recovery channeling pathways. Addressing the practical constraints of commingled production and injection in long intervals and the lack of production or injection profile data in offshore fields, a novel reservoir engineering method was developed for allocating production to individual sand bodies and determining their water-flooded patterns. This led to the creation of the Intelligent Diagnosis Technology for Water Flooding Status in Complex Reservoirs (FLIIS™), along with its corresponding software system (FLIIS). FLIIS assisted reservoir engineers in conveniently and efficiently evaluating the water-flooded status of individual sand bodies in target fields. It significantly enhanced workflow efficiency and reduced geological and reservoir risks associated with well intervention measures such as acidizing, fracturing, water shutoff, and profile control,as shown in Figure 1. FLIIS was also applicable for identifying thermal recovery channeling pathways in offshore heavy oil reservoirs. It enabled the precise characterization of channeling pathway dimensions, providing critical input for subsequent high-temperature profile modification and water shutoff operations.

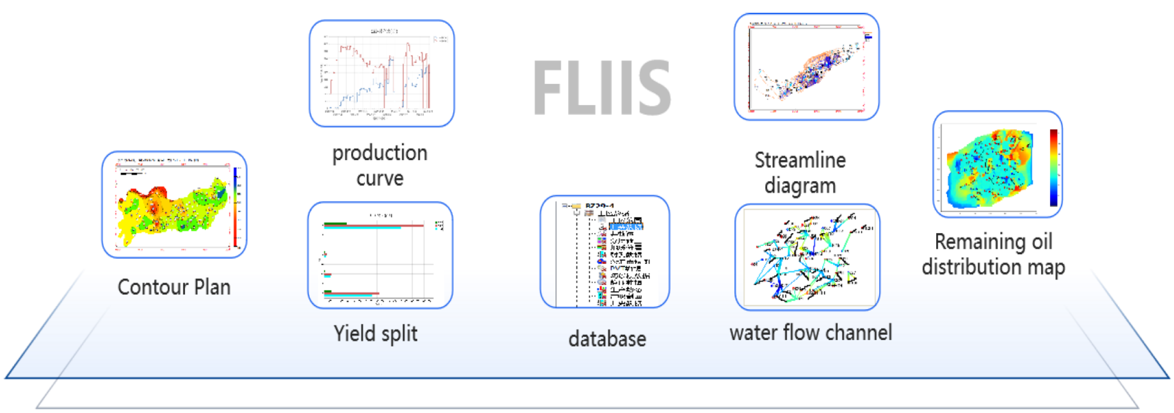

Figure 1—the workflow of FLIIS

Step 2: customization of water shutoff agents. Based on the characteristics of thermal recovery channeling in offshore heavy oilfields, static and dynamic laboratory studies were conducted to develop reversible thermosensitive gels and foam systems suitable for offshore heavy oil reservoirs. Compared to conventional gel polymers, thermosensitive reversible gels were dissolved in water at low temperatures. During thermal injection, as temperature raised, hydrophobic groups on the polymer chains became exposed, causing the polymer to gradually become insoluble. Through hydrophobic association, the polymer chains associated and coiled, forming a network structure that trapped water and gels within the pore spaces. This gel plugged high-permeability streaks, forcing injected fluids to divert and achieving the goal of profile modification and water shutoff. During production, as temperatures decrease, the polymer chains contracted, releasing water and re-dissolving. Foams utilized the Jamin effect to effectively plug high-permeability zones, characterized by their ability to "plug large pores, not small ones." When combined with nitrogen foam, the gel exhibited significant diversion effects, directing more gel-foam into high-permeability layers. This substantially increased the resistance factor in high-permeability zones while decreasing it in low-permeability zones, resulting in prominently improved profile modification effectiveness compared to conventional gels. Furthermore, foams can plug areas un-swept by the gel, adjust injected fluid viscosity, and expand sweep efficiency. Simultaneously, foams defoamed upon encountering oil. The gas released upon defoaming occupies pore space, partially enhancing formation energy, which aided in boosting well productivity after huff-and-puff cycles.

Step 3: optimized implementation of process parameters and dynamic control technology.The identification results of inter-well dominated flow channels reveal the development of these pathways between injector and producer pairs during steam flooding. These channels can be quantitatively described using inter-well connectivity coefficients and the equivalent radius of the channeling pathways. Based on experimental data from the selected plugging and profile control system, its performance can be characterized within numerical simulation software. Using the cost-to-production ratio and incremental oil per cycle as objective functions, the incremental oil production for different agent volumes can be forecasted. This process ultimately determines the optimal agent volume and the expected incremental oil gain. During field injection, downhole fiber optic monitoring was employed to track bottom-hole temperature changes, ensuring operational safety during agent placement. This data also provided direct input for future analysis of downhole and reservoir temperature fields in thermal injection wells. Prior to main agent injection, formation injectivity index tests were conducted to determine the minimum pressure required to initiate flow into the reservoir. During agent squeezed operations, the treatment pressure was strictly controlled to remain below this reservoir initiation pressure. This ensured the agent primarily entered the large pore channels, thereby minimizing formation damage to the lowest possible extent.

Thermosensitive reversible gel

Thermosensitive reversible gels are a class of smart materials that undergo a reversible physical change between a liquid-like state and a solid-like gel state (gel) in response to changes in temperature. The temperature sensitive reversible gel white powder product was obtained through the related preparation process.To compare the gelation performance of gels at different concentrations, experiments were conducted to investigate the effect of concentration on gel properties. Gel solutions with varying concentrations were poured into graduated cylinders to observe their flow behavior, and their viscosity was measured using a rotational viscometer. The comparative viscosity results for gels at different concentrations and temperatures are presented in Table 1.

Table 1—Viscosity comparison of gel at different concentrations and temperatures

T /°C	solution viscosity/ (mPa·s)		
	1.0%	1.5%	2.0%
30	100	1570	17600
40	71. 4	978	12800
50	46. 8	648	10831
70	≥3×10^4	≥3×10^4	≥3×10^4

The gel solution with a 3% mass fraction exhibits high viscosity and poor flow-ability,which may bedifficult to be pumped down-hole. Therefore, it is recommended to use a mass fraction ≤2%,as shown in Table 1, prepared with water at 40-50°C.

In order to better evaluate its performance, the gelation and failure temperature of thermosensitive reversible gels were measured. The optimal thermosensitive reversible gel solution with 2% mass fractionwas placed in ovens at 65°C, 70°C, 75°C and 80°C, respectively, and allowed to stand for a total of 12 hours to observe its gelation behavior. The gelation status at different temperatures was presented in Table 2. The results indicate that at 65°C, the gel formed a flowable jelly-like state with a yellowish color. At 70-75°C, the gel showed signs of gelation, but the effect was moderate. At temperatures more than 80°C, the gelation effect was excellent, resulting in a non-flowable gel with strength reaching Grade H. Consequently, the minimum effective gelation temperature for the thermosensitive reversible gel was determined to be 80°C.

Table 2—Gelling state of thermosensitive reversible gel at different temperatures

item	high-temperature treatment	gel forming temperature/°C	adhesive state (strength)	status
1	Not subjected to high temperature treatment	65°C	high deformed whole gel（F）	
2		70°C	moderately deformed whole gel（F）	
3		75°C	slightly deformed whole gel （H）	
4		80°C	slightly deformed whole gel （H）	

The 2% gel solution was prepared and aged at 160°C, 170°C, and 180°C for 12 hours. After retrieval, gel formation was observed at 75°C. The gel aging status at different temperatures was summarized in the table 3. The test results demonstrate that after aging at 160°C and 170°C, the gel successfully formed after cooled to 75°C.However, after treatment at 180°C, the gel failed to form upon temperature restoration to 75°C, indicating decomposition. The decomposed gel became a low-viscosity, homogeneous aqueous solution, free of precipitates or impurities, posing no risk of formation damage. Therefore, the deactivation temperature of the thermosensitive reversible gel was determined to be 180°C.

To evaluate the plugging performance of the thermosensitive reversible gel, one-dimensional thermal cycling permeability recovery tests were conducted under both high and low temperatures. The one-dimensional physical simulation oil displacement experiment employed different combinations of temperature, gas-liquid ratio, and chemical agent concentration. The steam/hot water injection rate was equivalent to the actual flow rate of fluid injected on-site. Since the specific heat and enthalpy of water vary significantly with temperature and pressure, they were determined according to the injection conditions[8].

Table 3—Gelling state of thermosensitive reversible gel at different temperatures

item	high-temperature treatment	gel forming temperature/℃	adhesive state (strength)	status
1	160℃	75℃	slightly deformed whole gel (H)	
2	170℃	75℃	slightly deformed whole gel (H)	
3	180℃	75℃	flowing gel (C)	

Step1: Core preparation andpermeability measurement were complished. The simulated core tube (Φ25 mm × 52 cm) was packed with simulated formation sand. Its permeability was then measured via water flooding.

Step 2: Injection of Gel. The prepared 1.5% gel solution was loaded into an intermediate container,due to core tube size limitations and accounting for potential losses during field injection. Using the intermediate container and a pump, the gel solution was injected into the core tube until saturated.

Step 3: The temperature was raised to 80°C and held constantly for 1 hour to allow gelation. Subsequently, water flooding was performed. The pressure differential was recorded, permeability was measured, and the plugging rate was calculated.

Step 4: The temperature was separately decreased to 55℃ and increased to 200℃. After holding each temperature constantly for 12 hours, water flooding was performed again. The pressure differential was recorded, permeability was measured, and the permeability recovery rate was determined.

The plugging capability of the high-temperature plugging agent was evaluated by physical simulation experiments. Quartz sand was cleaned, dried, and packed into sand-packed columns to create simulated artificial cores with permeabilities ranging from 11,000 to 13,000 mD , which targeted the permeability of 12,080 mD in the high-permeability zones of the reservoir. Displacement experiments were then conducted, using breakthrough pressure to characterize the system's plugging capacity.

Experimental results demonstrated that after aging and gelation the maximum displacement pressure was 4.05MPa as shown in Figure 2, which thermosensitive reversible gel solution formed a non-flowable and strong gel.After hot water displacement of 4PV which the time was230min, the sealing differential pressure was still 2.95mpa, and the sealing rate was greater than 99.7%,indicating its strong plugging ability under thermal recovery conditions.

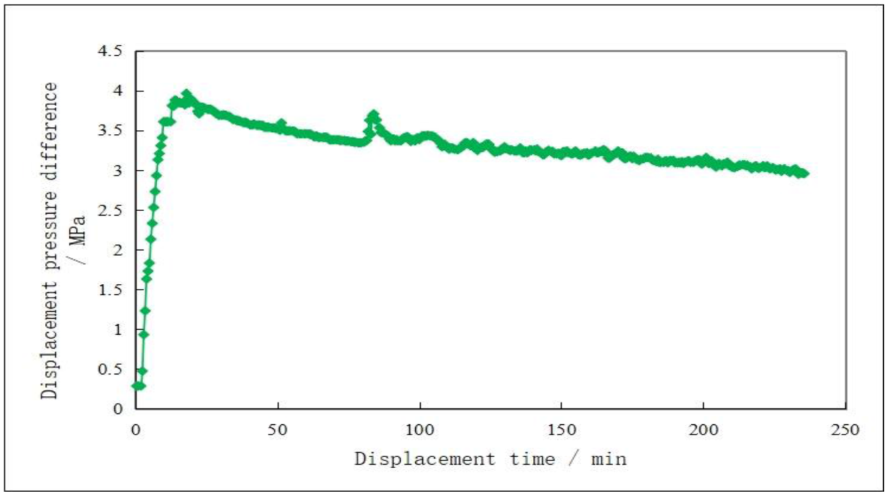

Figure 2—the curve of displacement

Leveraging the gel's responsive nature, its fluidity can be restored either by cooled with water or by applied high temperature as shown in Table 4. When the temperature was reduced to 55°C, the permeability recovery rate reached 87.08%. This occurred because the temperature fall below the gelation point, causing the majority of the injected gel to regain fluidity. After exposure to 200°C, the permeability recovery rate increased to 93.8%.This higher recovery rate resulted from the degradation of the gel induced by the high-temperature treatment.The thermosensitive reversible gel was an intelligent, reversible system characterized by its ability to provide "effective plugging without permanent damage",which was very friendly for engineering application.

Table 4—The blocking and degradation ability of the thermosensitive reversible gel

item	before gel injection Permeability K /mD	penetration rate after 2 hours of gelation at 80 °C k /mD	blocking rate /%	12h constant temperature water drive permeability k /mD	constant temperature T /°C	penetration recovery rate / %
1	13080	39	99.7%	11390	55	87. 08
2	11770	12	99.8%	11400	200	93.8%

Thermo-stable Foaming Agent

The thermo-stable foaming agent is a chemical additive designed to generate and stabilize foam structures at elevated temperatures without significant degradation or loss of function. Unlike conventional foaming agents that may break down under heat, thermo-stable variants maintain their foaming efficiency, gas release, and foam integrity in high-temperature environments. Considering the significant impact of temperature on foaming agent performance, high-temperature resistant foaming agents must be selected for thermal recovery applications. The results of the foam scan tested at 150°C and the resistance factor measurements at 250°C were presented in Table 5. The results indicated that while the foam half-life significantly shortened with increasing temperature, the resistance factor increased markedly. Following a comprehensive evaluation of performance metrics, including half-life and resistance factor, Foam-1 was selected as the stable foaming agent for thermal recovery operations.

Table 5—Comparison of foam scanning experiment (150 #) and resistance factor (250 #) measurement results

item	foam scanning test at 150 °C			resistance factor at 250 °C
	foaming time/s	stabilization time/s	half-life time/s	
Foam-1	40.23	169.95	240.09	39.31
Foam-2	41.14	53.00	164. 04	10.77
Foam-3	43.91	133.55	147.92	13.85
Foam-4	38.25	80.31	167.39	16.08
Foam-5	85.31	6.15	84.94	23.08

The performance ability was also measured by thermo-stable foaming Volume Scan. The red curve represented the foam volume height curve as shown in Figure 3. The time taken for the foam volume to increase from 0 to approximately 200 cm³ was defined as the foaming time. The duration during which the foam volume remains stable above 200 cm³ is defined as the foam stabilization time. The time required for the foam volume to decay to 100 cm³ was defined as the half-life. By analysis from the curve, the thermo-stable foaming agent Foam-1 exhibited that the foaming time was 40.23 seconds, stabilization time was 169.95 s,and half-life was 240.09 s.

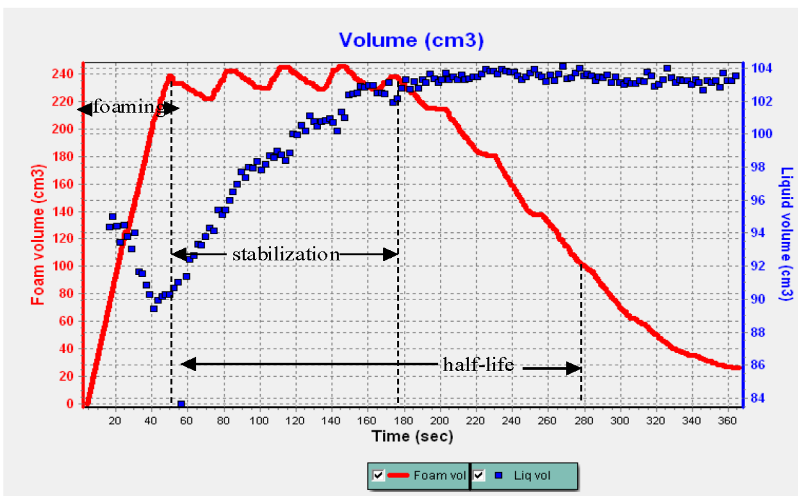

Figure 3—Scanning curve of thermo-stable foaming (150 #)

In order to better evaluate the high temperature profile modification and plugging capability of thermo-stable foaming agent at 250 ℃，the parallel core tube thermal displacement apparatus was employed by the experimental scheme in Table 6. The experimental procedure was as follows:

Step1: Prepare two simulated sand pack experiment.The one was low-permeability tube and the other one was high-permeability tube, with the permeability contrast of 1:4. It was to be accomplished to measure their permeability via water flooding and record the displacement differential pressure.

Step 2:Saturate the sand pack experiment with oil, then perform thermal displacement until the water cut reached 98%.

Step 3:Load the prepared foaming agent solution at 1%concentration into an intermediate container.

Step 4: Prepare an N_2/CO_2 mixed gas cylinder with a 4:1 ratio.

Step 5:Place the sand pack experiment into the displacement equipment. It was to be accomplished to set the temperature to 250°C and pressure to 2 MPa. Then it was to open the valves of the intermediate container and gas cylinder. Subsequently the injection of the foaming agent solution and gas was at gas-to-liquid ratio of 2:1, performing steam foam flooding.

Step 6:Stop foam injection and switch to water flooding after 2 hours of displacement.

Table 6—Experimental parameters of high temperature foam plugging experiment

sand pack	porosity/%	gas-measured permeability/10^{-3}um^2	Saturated oil volume/ml	initial oil saturation /%	Irreducible Water Saturation/%
Low permeability pipe	35.34	1675	21.03	83.11	16.89
High permeability pipe	40.3	6482	25.90	87.25	12.75

Figure 4 presents the experimental results evaluating the high-temperature conformance control capability of the thermo-stable foaming agent.

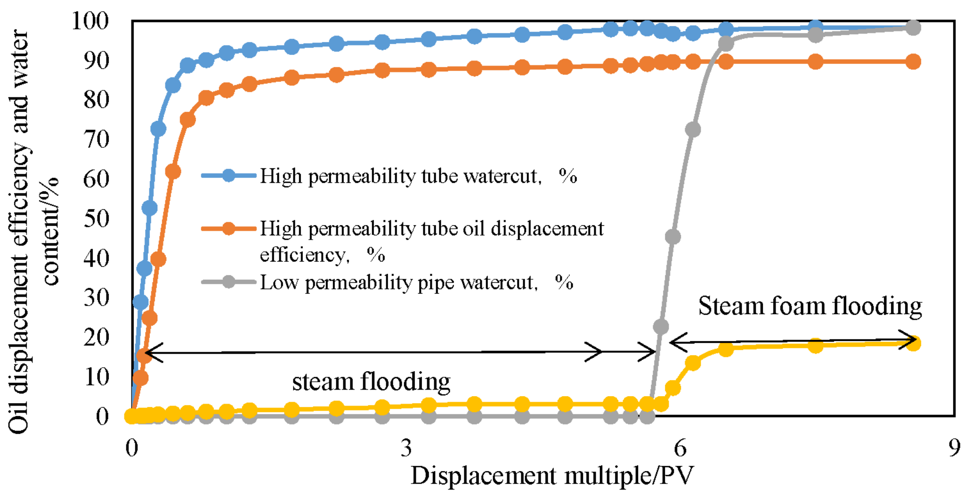

Figure 4—Evaluation of high-temperature profile control and displacement ability of thermo-stable foaming

Key findings from the experiment was the conformance control performance of high-temperature foam under heterogeneous conditions of permeability contrast 1:4. Steam flooding alone showed limited oil displacement in the low-permeability core section due to preferential channeling. Subsequent foam injection achieved15.31% increase in oil displacement efficiency in low-perm section, while 5–10% reduction in water cut in high-perm section. The mechanism was attributed to foam-induced blocking and fluid diversion, which expanded steam coverage into previously underutilized low-permeability zones – optimizing thermal energy distribution across heterogeneous layers.

Application

The customized plugging strength technology for heavy oil has been deployed at scale in multiple thermal recovery fields, including NB-1, LD-1, LD-2, and LD-3. Particularly notable results were achieved at LD-1 and NB-1 oilfield, contributing significantly to Bohai's thermal production exceeding 1 million tons in 2024 and providing strongsupport for sustained offshore crude oil production growth.

Application of thermo-stable foaming agent

The LD-1 oilfield was China's first offshore steam huff-and-puff demonstration project. Its main oil-bearing zones were buried at depths of 1,500–1,600 m. The crude oil exhibits high viscosity, which was 5,441 mPa·s at 50°C on the surface, and that was 2,908.8 mPa·s reservoir viscosity. It was an edge and bottomwater heavy oil reservoir with strong aquifer support, characterized by a water volume 4-11 times that of the oil volume.

As an edge and bottom-water heavy oil reservoir developed via steam huff-and-puff, it had a strong pressuredepletion process, which the LD-1 field was prone to rapid reservoir pressure decline, leading to edge or bottomwater influx. From 2022 to 2024, several thermal production wells in the LDoilfield experienced high water cut towards the end of their first cycle. After study, the technique by utilizing thermo-stable foaming agent was employed to suppress water coning and modify the steam injection profile.To date, this technique has been applied in 13 well treatments, resulting in cumulative incremental oil production exceeding 110,000 cubic meters. For these wells the initial average daily oil production increased by 3-5 times, meanwhile the water cut decreased by 50-60%. Moreover,most wells maintained effectiveness for over 300 days.

Taking well A1in the first cycle for example, the well experienced high water cut due to edge and bottom-water encroachment. Following implementation of the thermo-stable foaming agent process, the Initial water cut dropped from 74% to below 5%.Peak daily oil production increased from 4.9 m^3 to 53.9 m^3.Cumulative incremental oil reached 4,100 m^3.The effective period lasted approximately 5 months.

Application of thermosensitive Reversible Gel

The NB-1 oilfield was situated in the central Bohai Sea, with an average water depth of 12.2 m. In the southern area the density of surface crude oil was 0.941–0.978 g/cm^3, and the viscosity of surface crude oil was 929–4,946 mPa·s.The resin and asphaltenecontent were high with 21.04%–27.60%. The reservoir exhibits high porosity and permeability characteristics, which the porosity was 32.6%～36.1% with an average value of 35.1%, and the permeability was $2,179.9\times10^{-3}$～$5,626.2\times10^{-3}$ $\mu m^2$with an average value of$4,564.4\times10^{-3}$ μm^2.

NB-1 oilfield was put into operation in 9th, 2005. The north area was developed via water flooding. At present, there were 8 water injection wells. However, for the southern area it initially relied on conventional development by natural energy, with only one weak gel pilot injection well converted in May 2008. Since 2010, pilot thermal recovery trials for heavy oil had been conducted in the main sand bodies of the southern area using MCTF stimulation. The technology by using thermosensitive reversible gelwas first applied in Wells B3 and B4. These wells faced a major challenge, which significant gas channeling during MCTF injection, disrupting production and causing the shutdown of 3 adjacent wells. Following the injection of thermosensitive reversible gel, gas channeling during steam injection in individual wells was effectively suppressed. After thermosensitive reversible gelmeasure the oil production increased by 40-60%.

Table 7—Change of thermal recovery production before and after measures

Well	Daily production before gel injection, m^3/d	Daily production after gel injection, m^3/d
B3	23.26	50.5
B4	32.06	46.5

Conclusions

(1) This technology enables flexible selection of products and plugging techniques tailored to specific reservoir thermal recovery processes. The FLIIS™ quantitatively identifies dominant flow channels during heavy oil thermal recovery, providing the basis for customized profile modification and plugging strategies.

Furthermore, leveraging insights into channeling characteristics and size, two plugging agents with distinct strengths were developed, including a high-temperature resistant foaming agent and a thermosensitive reversible gel.

(2) The thermosensitive reversible gel was developed, featuring temperature-controlled gelation and reversible properties. Below its phase transition point of 80°C, it exists as a low-viscosity fluid. At above 80°C, it rapidly forms a colloid achieving a gel strength of Grade H, with a temperature resistance of 180°C. The formulation can be adjusted to suit different reservoir temperatures. The plugging efficiency exceeds 99.7%. Crucially, its reversible nature allows blockage removal via cooling or heating in case of accidental plugging. Field applications in NB-1 field's MCTF huff-and-puff operations demonstrate its effectiveness in mitigating gas channeling interference, resulting in a 40-60% increase in post-stimulation oil production.

(3) The thermo-stable foaming agent resistant to temperatures up to 350°C was developed, exhibiting an oil resistance of 30%. It maintains a high resistance factor even after aging, effectively inhibiting steam channeling during multiple thermal recovery cycles and enhancing development efficiency. Field application results in LD-1 fieldillustrate its impact, which immediately post-treatment using the synergistic thermal-agent-gas process, the water cut dropped from 74% to below 5%. Daily oil production peaked at 53.9 m^3 , yielding a cumulative incremental oil production of 4,100 m^3 over an effective period of approximately 5 months.

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